

Feedback and Newcomer Retention in X Community Notes

CS 5306 / INFO 5306, Project 1

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Abstract

In X’s Community Notes, a contributor’s first note can be rated Helpful, rated Not Helpful, or sit unresolved in “Needs More Ratings” indefinitely. We ask whether an explicit rejection predicts departure differently from never receiving a verdict at all, a comparison the Wikipedia newcomer literature cannot make. Using a full public snapshot of 3.1 million notes and classifying each first note by the status it held seven days after it was written, we find that contributors rejected within seven days were 15.0 percentage points less likely to write again within thirty days than contributors whose note was still unresolved (95% CI -15.5 to -14.5). That figure is mostly not discouragement. In April 2024 X began revoking writing ability after a single Not Helpful note, and the gap triples across that date, from 6.1 to 17.1 points; 89.4% of rejected first-time authors have since lost writing ability, typically the same day the verdict arrived. Cohorts predating the rule, who could not be locked out, still show a gap of about five points. We report that residual as the defensible estimate, and the pooled figure as an artifact of a design decision the platform made.

1 Introduction

Crowdsourced moderation systems depend on newcomers who stay. Community Notes asks volunteers to write short corrections on posts, then has other volunteers rate them. A matrix-factorization scorer publishes a note only when contributors who usually disagree agree it is helpful [4]. The system is deliberately hard to satisfy, and most notes never satisfy it.

That produces an outcome structure with no clean analogue elsewhere. A first note can be rated Helpful, rated Not Helpful, or receive no verdict at all, remaining in “Needs More Ratings” (NMR) for as long as it exists. Wikipedia’s newcomer literature has established that early negative feedback predicts departure: reverting a newcomer’s first edit measurably reduces the chance they edit again [1, 2]. But on Wikipedia an edit that is not reverted stands. There is no large population of contributors whose first contribution was met with silence in the way that most Community Notes contributions are.

Our question is whether explicit rejection predicts different retention than getting no verdict at all. Not Helpful versus NMR is the comparison; Helpful is a reference point. This matters for design, because a system whose modal outcome is “no decision yet” is making an implicit choice about how to treat newcomers, and it is not obvious whether silence is gentler than rejection or worse.

The answer turned out to depend on something we did not anticipate. Rejected contributors are far less likely to return — but for most of the period we study, rejection does not merely discourage a contributor, it mechanically strips their ability to write. Separating

the two is the substance of this paper, and Section 6.2 is where we do it. The short version: an explicit Not Helpful verdict predicts sharply lower continued contribution than no verdict, most of that gap is the platform’s writing lockout rather than discouragement, and a smaller association of roughly five points survives in cohorts that predate the lockout rule.

2 The System

Two mechanics shape everything that follows.

Contributors earn the ability to write by rating first. New contributors may only rate. They unlock writing by reaching a Rating Impact of 5, earned by rating notes before they reach a status and matching the status they go on to reach [6]. This requirement was introduced in September 2022 and applies to every cohort we analyse. *Our “newcomers” are therefore newcomers to writing only.* Each had already rated enough notes, accurately and early enough, to clear a quality bar. This weakens the analogy to Wikipedia newcomers, where no comparable filter exists, and it means our population is more committed than a typical first-time participant.

Not Helpful ratings can cost a contributor the ability to write. A contributor’s writing ability is locked when three of their last five resolved notes are Not Helpful — or, since 2 April 2024, when *any* note reaches Not Helpful while their Writing Impact is zero or negative [6]. Writing Impact starts at zero and only moves when a note resolves. A first-time author therefore has a Writing Impact of exactly zero, so from April 2024 onward a single Not Helpful first note locks writing on its own, while a first note sitting in NMR does nothing. The contrast this paper measures is a contrast the platform itself acts on. We return to this in Section 6.2.

A note’s status is not permanent, and statuses are recomputed periodically, but the public data records when each note first left NMR and what it became [7]. That first transition is what we use.

3 Related Work

The closest body of work is on Wikipedia newcomer retention. Halfaker et al. [2] show across 400,000 revisions that having an early edit reverted is powerfully demotivating, reducing the volume of subsequent work, while still improving aggregate quality — the system gains more from rejecting bad work than it loses by driving away the people who did it. The later account of Wikipedia’s decline [1] attributes much of it to increasingly automated and impersonal rejection of newcomers.

Community Notes differs in two ways that make it a useful contrast. Rejection is the output of a scoring rule rather than a person’s judgment, so a rejected contributor has no editor to argue with and no revert summary to read. And the modal outcome is neither acceptance nor rejection but silence, which Wikipedia

does not produce at comparable scale. Recent descriptive work covers the program’s first four years, including contributor activity and rating behaviour [3], but not, to our knowledge, the retention consequences of the no-verdict condition.

4 Data

We use the public Community Notes bulk downloads [5], snapshot dated **18 September 2026**. Each snapshot is cumulative back to 2021, so one is enough. Upstream keeps roughly a week of snapshots and older dates return 404, so the archive we took locally is the only copy; nothing in our pipeline writes to it.

Three tables: notes (3,086,867 rows), noteStatusHistory (3,290,294 rows), and userEnrollment (1,490,672 rows). The project rests on two fields in the second: firstNonNMRStatus, the first real verdict a note received, and timestampMillisOfFirstNonNMRStatus, when it arrived. Both are empty when a note never left NMR, which is true of 82.7% of notes.

Snapshots contain only notes created up to 48 hours before release, so the effective data cutoff is **16 September 2026**, two days before the snapshot date. All censoring uses the cutoff, not the snapshot date.

We did not use the ratings table, which runs to tens of gigabytes and answers a different question — we need note outcomes, not individual ratings — or the text of the annotated posts, which is not public.

Author identifier stability. Our analysis assumes a participant identifier means the same person across releases. Because snapshots expire, this can only be checked while a second one is still downloadable, so we did it: comparing noteStatusHistory across the 18 and 16 September releases gives 3,282,773 notes present in both and **zero** author-identifier mismatches.

Two data problems we found rather than assumed. First, firstNonNMRStatus is not merely sparse before May 2022, it is *empty*: not one of the 28,988 notes created before then has a recorded verdict. Those cohorts are 100% NMR by construction, not by outcome. Second, 233,514 notes (7.6%) have no noteStatusHistory row at all, spread across 28 months from March 2024 onward at a rate rising from 6.7% to 14.1%. Those notes have *no recorded outcome*, which is not the same as never leaving NMR; treating them as NMR would inflate the comparison group by a growing amount and manufacture a cohort trend. Both drive exclusions in Section 5.4.

5 Method

5.1 Contributor-level dataset

We join notes to noteStatusHistory on note identifier and emit one row per author: their first note, its outcome and timing, their next note, their cohort month, and first-note features (classification, whether it cites a URL, length, reason tags). A first note is the author’s earliest note in the snapshot, ties broken by note identifier so the ordering is total and the build deterministic. Since snapshots are cumulative back to 2021, that is genuinely their first note. The table has 343,176 rows, one per distinct author in notes.

Before using it we validated it against the raw files: twelve structural checks, plus twenty randomly sampled authors rebuilt field by field from the raw TSVs by a separate Python parser that does

not go through our query engine. All twelve checks and all twenty authors matched.

5.2 Landmark classification

This is the central methodological choice. **We classify each first note by the status it held seven days after it was written**, not by the status it eventually reached.

Classifying by eventual status uses information that did not exist when the contributor decided whether to come back. A note that sat in NMR for three months and then turned Not Helpful was, for the entire period in which the contributor was deciding, a note with no verdict. Worse, the eventual-status NMR group is partly a function of when the snapshot was taken rather than of anything the contributor experienced.

Seven days is generous. From 2023 onward, the median first note reaches a verdict in under a third of a day and the ninetieth percentile inside two days, so a seven-day landmark captures nearly every verdict that will ever arrive. We also drop authors who wrote again *before* day seven: no feedback could have reached them, so their return says nothing about how they reacted to one. A verdict landing after day seven counts as NMR here, by design.

5.3 Retention and censoring

Retention is whether the author wrote another note between the landmark and the end of a 30-, 60-, or 90-day window. An author enters a window only if the entire window closes before the data cutoff, so each window has its own sample size. Proportions carry Wilson 95% intervals and differences carry Newcombe hybrid-score intervals.

Section 6.3 reports Kaplan-Meier curves instead, which censor rather than discard, as a check that the windowing is not doing the work.

5.4 Exclusions

Exclusions are boolean columns applied in one place, never deleted rows. Of 343,176 authors we keep 322,094:

- 5,052 authors in cohorts before January 2023, where verdicts are absent or were rewritten by a January 2023 rescoring of all historical notes.
- 15,997 authors whose first note has no status-history row, whose outcome is unknown rather than absent.
- 35 AI note-writer accounts, identified by an enrollment state prefixed `api`. They are negligible as authors but wrote 8.6% of all notes, and an AI that stops writing has not given up.

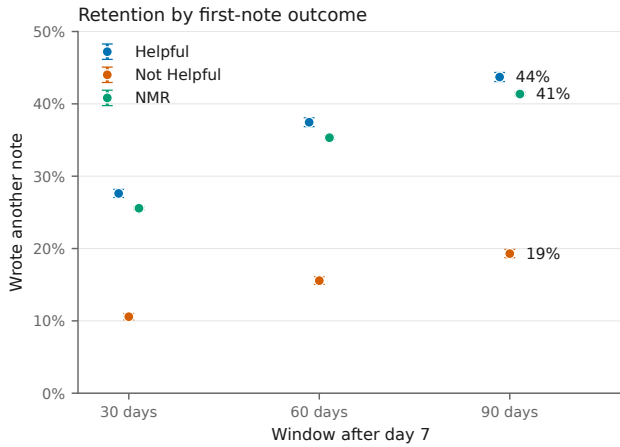
Applying the landmark leaves 230,588 authors observable at day seven, of whom 227,324 have a fully observed 30-day window.

6 Results

Contributors whose first note was rated Not Helpful within seven days were 15.0 percentage points less likely to write another note in the following thirty days than contributors whose first note was still unresolved (95% CI −15.5 to −14.5; $n = 19,153$ Not Helpful and 183,373 NMR). Section 6.2 explains why that number is mostly not what it appears to be.

Table 1: Retention by first-note outcome at day 7. Wilson 95% intervals.

First-note outcome	<i>n</i>	30-day	90-day
Helpful	24,798	27.6%	43.7%
Not Helpful	19,153	10.6%	19.3%
No verdict (NMR)	183,373	25.6%	41.4%

**Figure 1: Retention by first-note outcome at 30, 60, and 90 days, with 95% intervals.**

6.1 The Not Helpful versus no-verdict comparison

Table 1 and Figure 1 show the answer to the paper’s question, and the shape is asymmetric. Rejection is associated with a large deficit: -15.0 points at thirty days, widening to -22.1 at ninety. Approval is associated with almost nothing: $+2.1$ points at thirty days and $+2.4$ at ninety. Getting no verdict at all is, on these numbers, nearly as good as being told your note was helpful.

The specification matters more than we expected. Computed the obvious way — labelling each note with whatever verdict it eventually received and measuring retention from the note itself — the same data gives -21.1 points, overstating the difference by 6.1 points. Most of that comes from keeping the authors who wrote again within the week, the most engaged people in the sample, without asking whether any feedback had reached them. We hold censoring identical between the two so the comparison isolates the landmark. That the easy specification is wrong by forty percent is itself worth reporting.

6.2 Cohorts, and a structural break

Splitting by cohort month changes the finding. Figure 2 shows the gap is small and often indistinguishable from zero through 2023 and early 2024, then collapses. Not Helpful retention falls from around 28% in February 2024 to 9.6% in April 2024 and stays in a 3–8% band thereafter, while the other two groups drift down together and far more gently. Pooled either side of that date, the Not Helpful minus

NMR gap is -6.1 points before April 2024 (95% CI -7.4 to -4.7) and -17.1 points after (-17.6 to -16.7).

April 2024 is when X began locking writing ability on a single Not Helpful note (Section 2). The enrollment data confirms the mechanism directly: 89.4% of **rejected first-time authors have lost writing ability at least once**, against 16.8% of the no-verdict group, and 48.7% are still locked out. Among rejected authors who wrote exactly one note and have been locked out, the median gap between the verdict arriving and the lockout is 0.03 days and 84.0% were locked within a week. The verdict and the lockout are effectively the same event.

This does not reduce the finding to zero. The enrollment field records only a contributor’s most recent lockout, so a 2023 author’s could have occurred years later under the new rule. Counting only lockouts stamped before the rule date, just 1.2% of pre-rule rejected authors had lost writing ability when it mattered, against 0.3% of the no-verdict group. Those cohorts really were free to keep writing, and they still show a 6.1-point gap. Even if every locked contributor would otherwise have returned, at most 1.2 points of that is mechanical, leaving **roughly five points that is not**.

We therefore report two numbers with different jobs. The pooled -15.0 points describes what happens to rejected newcomers in Community Notes today, and is mostly the platform’s lockout doing what it was designed to do. The -4.9 points from pre-rule cohorts is our estimate of the association between rejection and departure absent that mechanism.

6.3 Time to return

Kaplan-Meier estimates agree with the windowed proportions and rule out the windowing as an artifact. Follow-up starts at the day-7 landmark, the event is the author’s next note, and authors who never return are censored at the data cutoff rather than dropped. By day ninety 43.7% of Helpful authors, 41.2% of NMR authors and 19.0% of Not Helpful authors have returned, within half a point of the windowed figures in Table 1. No group reaches a median time to return within ninety days — most contributors never write a second note at all.

6.4 Adjusted estimates

A logistic regression of 30-day retention on day-7 status, with cohort-month fixed effects and first-note controls (classification, URL presence, length, and reason tags), gives an odds ratio of 0.36 for Not Helpful (95% CI 0.34–0.38), an average marginal effect of -18.5 points. That is 3.5 points larger than the raw difference, not smaller. The reason is Section 6.2: cohort fixed effects make the comparison within cohort, which removes the early cohorts’ smaller gaps from the average. We report both rather than the more convenient one. One control is worth noting: a first note asserting a post is *not* misleading is associated with $+2.5$ points of retention once status is held constant, despite such notes being structurally ineligible for Helpful status and reaching Not Helpful far more often.

7 Robustness

Table 2 reruns the primary comparison under each variation. The difference is negative in all six, ranging from -5.7 to -18.2 points

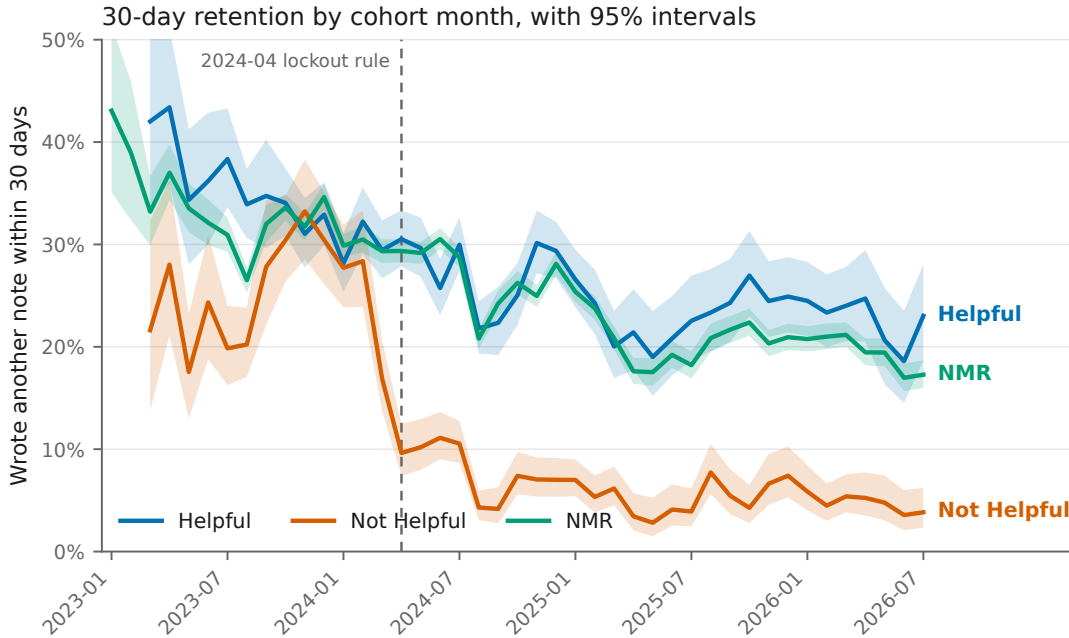


Figure 2: 30-day retention by cohort month and first-note outcome, with 95% intervals. The dashed line marks April 2024, when a single Not Helpful note began locking writing ability. Cohort months too recent for a 30-day window to close before the data cutoff are omitted.

Table 2: Not Helpful minus NMR difference at 30 days under each variation.

Variation	Difference
Baseline (day-7 landmark)	-15.0 pp
Landmark at 3 days	-18.2 pp
Landmark at 14 days	-11.7 pp
Excluding not-misleading first notes	-14.0 pp
Retention = two or more later notes	-5.7 pp
Cohorts with 90+ days after the window	-15.0 pp
Naive eventual-status specification	-21.1 pp

against a baseline of -15.0, with no sign flips. Moving the landmark trades sample composition against classification accuracy in the expected direction. The stricter retention definition compresses the difference because base rates fall for every group. Nothing here is close to the sensitivity that matters, which is the cohort split.

8 Threats to Validity

Writing lockouts. The largest threat, and the one that reshaped the paper. We have bounded it in Section 6.2 rather than dismissed it, but the bound is one-sided. `userEnrollment` carries one current state per contributor with no history, so after April 2024 we cannot separate a contributor who was locked at the moment they would have written from one who had already lost interest. An author who was locked, earned back in, and left anyway is indistinguishable

from one never locked. Our pre-rule estimate sidesteps this by using a period when the mechanism did not exist, at the cost of a smaller and older sample.

Selection. We did not assign anyone to Not Helpful or NMR. A weak first note plausibly produces both a bad verdict and a contributor who was leaving regardless. Controlling for observable note features is not the same as solving this, and we claim association throughout.

Notification. We do not observe whether a contributor ever saw their note’s status. The landmark assumes seven days is enough for feedback to arrive, which is a claim about the system’s notifications that we cannot verify from public data.

Survivorship. Deleted notes disappear from notes while remaining in `noteStatusHistory`, so our author population is drawn from surviving notes only.

Automated authors. AI note writers entered the system in 2025 and are excluded, but only via current enrollment state; an account since moved to another state would be invisible.

9 What We Did

Four weeks, three people, and the work was mostly not the statistics.

The first week went on establishing what the system actually does, which turned out to be the highest-value thing we did. Reading the scoring source alongside the published guide is what surfaced the April 2024 lockout rule and, critically, its *date* — which is what later let us separate the mechanism from the behaviour instead of writing a paragraph apologising for the confound. The guide’s rendered pages are client-side rendered and return nothing to a

plain fetch, so we worked from the markdown sources the site is built from.

The second week went on the data, and produced two stops. We found that first-verdict fields are entirely empty before May 2022, which killed the 2021–2022 cohorts outright, and that 7.6% of notes have no status-history row at all, which is undocumented and would have silently inflated our comparison group at a rate that grows over time. Both were flagged and escalated rather than patched around, which cost a day each and was worth it. We also verified author-identifier stability across two releases inside the one-week window during which that check is possible at all.

Infrastructure bit in small ways worth recording. The Python environment is a pinned Nix flake; an initial attempt with uv failed because its downloaded interpreter is dynamically linked against a loader that does not exist on NixOS. The survival library we planned to use does not build against the pinned package set, so Kaplan-Meier and the log-rank test are computed with `statsmodels` instead. Loading the tables with default CSV quoting silently corrupts rows, because note text contains bare double quotes.

Two mistakes we caught ourselves. Drawing the cohort figure revealed that the most recent month spikes in every series, because only authors who joined in the first days of that month can have a 30-day window close before the cutoff — a biased slice, not a cohort. We fixed it in the analysis with an explicit completeness flag rather than trimming the chart. And our first pass at the regression writeup claimed the adjusted and raw estimates agreed, when they differ by 3.5 points; the honest explanation of *why* they differ turned out to be more informative than the agreement would have been.

Reproducibility was a constraint throughout rather than an afterthought. The contributor table was frozen after validation and never rebuilt. Exclusions live in one function. Every number in this paper is generated into a single JSON file that the paper quotes, so no figure here was typed by hand.

10 Conclusion

An explicit Not Helpful verdict on a first note predicts substantially lower continued contribution than no verdict at all — 15.0 points at thirty days — but most of that is X’s writing lockout rather than contributors choosing to leave. In cohorts predating the lockout rule, where rejection carried no mechanical penalty, the association is about five points. Rejection discourages, modestly; the platform then removes the option entirely, which is what the pooled number mostly measures.

The result we did not expect is the asymmetry. Approval buys almost nothing: contributors told their note was Helpful returned only about two points more often than contributors told nothing at all. For a system whose modal outcome is silence, that is reassuring in one way and troubling in another. Silence is not experienced as rejection, so the 82.7% of notes that never resolve are not quietly driving people away. But the system also gets very little retention benefit from telling someone they did well. If Community Notes wants newcomers to stay, the lever is not faster verdicts — it is reconsidering whether a first rejection should cost someone the ability to try again, given that 89.4% of rejected first-time authors lose it, usually the same day.

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